

2. Determine whether f is a function from $\mathbb{Z}$ to $\mathbb{R}$ if

a) $f(n) = \pm n$ b) $f(n) = \sqrt{n^2 + 1}$ c) $f(n) = 1/(n^2 - 4)$

Solution: a) $f(n) = \pm n$ is not a function from $\mathbb{Z}$ to $\mathbb{R}$ because for each integer n , $f(n)$ assigns n to two values $+n$ and $-n$, which violates the definition of a function, where each element of the domain must be assigned to a unique element of the codomain.

b) $f(n) = \sqrt{n^2 + 1}$. This satisfies the definition of a function, since for each integer n , the function $f(n)$ is mapped to a unique real number $\sqrt{n^2 + 1}$. $(n^2 + 1) > 0 \therefore \sqrt{n^2 + 1} \in \mathbb{R}$. The function is defined.

c) $f(n) = 1/(n^2 - 4)$. $f(2) = 1/(2^2 - 4) = 1/0$. Similarly $f(-2) = 1/0$ is not defined. $\therefore$ It's not a function since it's not defined for the integers ± 2 . $\square$

3. Determine whether f is a function from the set of all bit strings to the set of integers if

a) $f(S)$ is the position of a 0 bit in S .

Solution: A string S ~~can contain~~ can contain more than 1 zero's.
Ex: $f(1001) = 2, 3$. This violates the definition of a function where each element in the domain is mapped to ~~exactly~~ ^{unique} one element in the co-domain.
 $\therefore f(S)$ is not a function.

b) $f(S)$ is the number of 1 bits in S .

Solution: A bit string consists only of 0's and 1's. It's intuitive that each string S has a unique number of 1's. This satisfies the definition of a function and hence $f(S)$ is a function.

c) $f(S)$ is the smallest integer i such that the i th bit of S is 1 and $f(S) = 0$ when S is the empty string, the string with no bits.

Solution: If the string S has no 1's, then $f(S)$ is not defined.
For example, if $S = 0, 00, 000, \dots$, $f(S)$ is not defined.
 $\therefore$ It's not a function. $\square$